higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ENGINEERING PROCESSES
NQF LEVEL 4

(6021004)

19 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 7 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only blue or black ink.
6. Write neatly and legibly.

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QUESTION 1

1.1 The choice of lubrication depends on the function it has to perform.

State FOUR factors that one needs to consider when selecting the correct lubricant. (4)

1.2 Define the following terms used in engineering:

1.2.1 Flashpoint

1.2.2 Viscosity

(2 × 1) (2)
[6]

QUESTION 2

2.1 When working in the workshop it is your responsibly to ensure that your working area is free of obstacles and that conditions are safe.

Name FIVE aspects that must be checked to ensure that the working area is free of obstacles and that conditions are safe. (5)

2.2 Name the cleaning method to apply when equipment is still in one piece. (1)

2.3 Replacement parts are inspected for quality and correctness.

State THREE reasons why replacement of components need to be recorded. (3)

2.4 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.4.1–2.4.10) in the ANSWER BOOK.

2.4.1 Life expectancy is the length of time that something is expected to continue to operate or be useful.

2.4.2 When you select and assemble parts, it is important to ensure that the component of the machine functions regardless of the time spent on the job.

2.4.3 Examining the component or its parts carefully will help you to determine what the problems are so that you can repair them and prevent it from recurring.

2.4.4 Manufacturers need to know what causes components to fail in order to help them to find ways to prevent or minimize faults on their machines.

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2.4.5 An artisan is a person who sits and works in the office.

2.4.6 Locknuts are used to lock another nut in position.

2.4.7 Spring washers absorb water and oil.

2.4.8 Studs are square rods threaded at both ends.

2.4.9 Cap screws are threaded all the way to the screw head.

2.4.10 Welding is a secure method of joining two pieces of metal permanently.

(10 × 1) (10)

2.5 Name the FIVE aspects that one needs to check when examining components for defects. (5)

2.6 Explain the following terms used in engineering:

2.6.1 Casting

2.6.2 Friction

(2 × 1) (2)

[26]

QUESTION 3

3.1 Choose a description from COLUMN B that matches a term in COLUMN A. Write only the letter (A–M) next to the question number (3.1.1–3.1.10) in the ANSWER BOOK.

COLUMN A		COLUMN B	
3.1.1	Reaming	A	provide movement for cutting
3.1.2	Chuck	B	the process of removing sharp edges
3.1.3	Boring	C	a metal transformed to a liquid by heating it at a very high temperature
3.1.4	Deburring	D	part of a machine that keeps something firmly so that it does not move
3.1.5	Feed screw	E	metal block used to press or cut something into a particular shape
3.1.6	Simulate	F	to remove something by manufacturing a hole
3.1.7	Transmit	G	the process of drilling holes and machining them to larger, accurate dimension
3.1.8	Edit	H	removing parts or mistakes that are unacceptable
3.1.9	Geometric	I	set of numbers which give the exact position of a point
3.1.10	Configure	J	to make or produce something that is not real but has the appearance of being real
		K	arrange something, especially computer equipment, so that it works with other equipment
		L	send out electronic signal or message
		M	having or using the shapes and lines in circles and squares

(10 × 1) (10)

3.2 Machines need to be monitored to ensure that the operational requirements are met.

State FOUR aspects that one needs to consider to determine whether the operational requirements of the machine are met. (4)

3.3 Explain how you will restore your work area of unnecessary clutter after completing a job. (2)

[16]

QUESTION 4

4.1 Which TWO elements combine to form alloy brass? (2)

4.2 What is meant by the *tensile strength* of a material? (2)

4.3 Various cutting processes are used to cut metal in an engineering plant.

State FOUR cutting processes that an artisan must consider when cutting material in an engineering plant. (4)

4.4 As part of your ISAT you are required to cut a work piece in the workshop.

State FIVE factors to consider when selecting a cutting machine. (5)

4.5 Cutting operations need to be checked on a piece of material before the cutting operation takes place.

State FOUR aspects that need to be checked before cutting operations start.

(4 × 2) (8)

[21]

QUESTION 5

5.1 Explain what a post processor is. (2)

5.2 What does the following abbreviations stand for?

5.2.1 CAM

5.2.2 APT

(2 × 1) (2)

5.3 Name the codes that are used to perform the following when manufacturing components using CAM:

5.3.1 To increase or decrease the feed rate

5.3.2 To control what type of motion or action is carried out

5.3.3 To sequence numbers for each block of instruction in the program

5.3.4 To instruct the commander which tools to use

5.3.5 For setting the spindle speed (5)

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5.4 Describe how you will define and verify component surfaces on drawn outlines. (2 × 2) (4)

5.5 Explain in detail how components can be examined for inaccuracies and also indicate how these can be corrected. (2 × 2) (4)
[17]

QUESTION 6

6.1 State FIVE economic reasons that one needs to consider when selecting joining methods. (5)

6.2 State FIVE aspects that one will check to determine whether the joining materials chosen conform to the required standard. (5)

6.3 Explain the meaning of the following terms used in engineering.

6.3.1 Durability

6.3.2 Composition

(2 × 2) (4)
[14]

TOTAL: 100

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